BASIC SQL COMMANDS

SQL STATEMENTS

SQL statements are classified as follows:

Data Retrieval Statement:

SELECT is the data extracting statement which retrieves the data from the database.

Data Manipulation Language (DML):

This language constitutes the statements that are used to manipulate with the data. It has three commands, which are INSERT, UPDATE and DELETE.

Data Definition Language (DDL):

This is the language used to define the structure of the tables. It sets up, changes, and removes data structures from the tables. It uses 5 commands, which are CREATE, ALTER, DROP, RENAME and TRUNCATE.

Data Transaction Language (DTL):

This is the language used to do undo and redo the transaction performed in the database. The commands are Commit, Rollback, and Save Point

Data Control Language:

This language is used to sanction the rights to the users to use the other user's database objects. The commands are Grant, Revoke

BASE SCHEMA

EMPLOYEE

Name	Type
EMPLOYEE_ID	NUMBER(3)
FIRST_NAME	VARCHAR2(10)
LAST_NAME	VARCHAR2(10)
MGR	NUMBER(4)
HIRE_DATE	DATE
JOB_ID	VARCHAR2(10)
SALARY	NUMBER(10)
COMMISION	NUMBER(8)
DEPTNO	NUMBER(2)

DEPARTMENT

Name	Type
DEPTNO	NUMBER(2)
DNAME	VARCHAR2(14)
LOC	VARCHAR2(13)

BONUS

Name	Type
ENAME	VARCHAR2(10)
JOB	VARCHAR2(9)
SAL	NUMBER(10,2)
COMM	NUMBER(10)

JOBGRADE

Name	Type
JOB_ID	VARCHAR2(10)
GRADE	NUMBER
LOSAL	NUMBER
HISAL	NUMBER

DATA TYPES IN ORACLE:

Data Type	**Description**
VARCHAR2(size)	Variable-length character data
CHAR(size)	Fixed-length character data
NUMBER(p,s)	Variable-length numeric data
DATE	Date and time values
LONG	Variable-length character data up to 2 gigabytes
CLOB	Character data up to 4 gigabytes
RAW and LONG RAW	Raw binary data
BLOB	Binary data up to 4 gigabytes
BFILE	Binary data stored in an external file; up to 4 gigabytes
ROWID	A 64 base number system representing the unique address of a row in its table

ORACLE 9I TABLE STRUCTURES

- Table can be created at any time
- No need to specify the size of table, the size is ultimately defined by the amount of space allocated to the database as a whole.
- Tables can have up to 1000 columns

NAMING RULES

Table names and Column names

- Must begin with a letter
- Must be 1-30 characters long
- Must contain only A-Z,a-z,0-9,_,$,#
- Must not duplicate the name of another object owned by the same user
- Must not be a reserved word

Data Definition Language (DDL)

1. Create 2. Alter 3. Drop 4. Truncate 5. Rename

1. a. Creating a table

Syntax:

Create table <Table Name>

(<Field1> <Data Type> <(width) <constraints> ,

<Field2> <Data Type> <(width)> <constraints>,

.....);

Example:

SQL> create table employee

(employee_id number(3),

first_name varchar2(10),

last_name varchar2(10),

mgr number(4),

hire_date date,

job_id varchar2(10),

salary number(10),

commision number(8),

deptno number(2));

Output:

Table created.

Example:

SQL> create table department

(deptno number(2),

dname varchar(14),

loc varchar(13));

Output:

Table created.

Note:

Other tables can be created in the similar way.

b. To view the Structure of the table, desc command is used

SQL> desc employee;

Name	Null?	Type
EMPLOYEE_ID		NUMBER(3)
FIRST_NAME		VARCHAR2(10)
LAST_NAME		VARCHAR2(10)
MGR		NUMBER(4)
HIRE_DATE		DATE
JOB_ID		VARCHAR2(10)
SALARY		NUMBER(10)

COMMISION	NUMBER(8)
DEPTNO	NUMBER(2)

2. Alter Table Statement:

Alter command is used to perform the following action on the table:

a. Adding column in the existing table
b. Increasing and decreasing the column size and changing data types
c. Dropping column
d. Renaming the column
e. Adding and dropping constraints to the table(discussed in constraints topics)
f. Enabling & disabling constraints in the table(discussed in constraints topics)

a. To Add a column to the table (structure)

Add option is used to add a new column

Syntax:

Alter Table <Table-Name> Add <Field Name> <Type> (width);

Example:

SQL> alter table employee add address varchar2 (20);

Output:

Table altered.

b. To Modify a field of the table

- Increase the width or precision of numeric column
- Increase the width of numeric or character columns
- Decrease the width of the column only if the column contains only null values or if the table has no rows
- Change the data type only if the column contains null values

Syntax:

Alter Table <tablename> MODIFY (<column name > < newdatatype>);

Example:

SQL> alter table employee modify address varchar2 (10);

Output:

Table altered.

c. To Drop a field of the table

Drop option is used to delete a column or remove a constraint

Syntax:

Alter Table <tablename> DROP COLUMN < column name>;

Example:

SQL> alter table employee drop column address;

Output:

Table altered.

d.To rename a column

Syntax:

ALTER TABLE <tablename> RENAME COLUMN <oldcolumnname> TO

<newcolumn name>

Example:

SQL> alter table employee rename column mgr to manager;

Output:

Table altered.

- **To Drop a table - Deletes a Table along with all contents**

Syntax:

Drop Table <Table-Name>;

Example:

Drop Table Student_table;

Output:

Table Dropped

- **To Truncate a table - Deletes all rows from a table ,retaining its structure**

Syntax: Truncate Table <tablename>

Example:

SQL> truncate table employee;

Output:

Table truncated.

g. To rename a table- Renames a table with new name

Syntax:

Rename <oldtablename> To <newtablename>

Example:

SQL> rename employee to emp;

Output:

Table renamed

<u>Data manipulation Language (DML)</u>

1. Insert 2. Delete 3. Update 4.Select

1 Insert command is used to load data into the table.

a. Inserting values from user

Syntax:

Insert into <tablename> values (val1,val2 …);

Example:

SQL> insert into department values(10,'accounts','chennai');

Output:

1 row created.

b. Inserting values for the specific columns in the table

Syntax:

Insert Into <Table-Name> (Fieldname1, Fieldname2, Fieldname3,..) Values (value1, value2, value3,..);

Example:

SQL> insert into department (deptno,dname)values(20,'finance');

Output:

1 row created.

c. ***Inserting interactively(Inserting ,ultiple rows by using single insert command)***

Syntax:

Insert Into <tablename> Values(&<column name1> , &<column name2> …);

Example:

SQL> insert into employee values(&empid,'&fn','&ln',&mgr,'&hdate','&job',&sal,
&comm,&dept);

Enter value for empid: 111

Enter value for fn: Smith

Enter value for ln: Ford

Enter value for mgr: 222

Enter value for hdate: 21-jul-2010

Enter value for job: J1

Enter value for sal: 30000

Enter value for comm: 0.1

Enter value for dept: 10

old 2: &comm,&dept)

new 2: 0.1,10)

Output:

1 row created.

Note: Column names of character and date type should be included with in single quotation.

- **Inserting null values**

Syntax:

Insert Into <tablename> Values (val1,' ',' ',val4);

Example:
insert into department values('101','',chennai);

Output:

1 row created.

2. To Delete rows from a table

Syntax:

Delete from <table name> [where <condition>];

Example:

a) TO delete all rows:

SQL> delete from department;

Output:

89 rows deleted.

b) conditional deletion:

SQL> delete from department where loc='chennai';

Output:

1 row deleted.

3. Modifying (Updating) Records:

a. Updating single column

Syntax:

UPDATE <table name> Set <Field Name> = <Value> Where <Condition>;

Example:

SQL> update department set loc='Hyderabad' where deptno=20;

Output:

1 row updated.

Note: Without where clause all the rows will get updated.

b. Updating multiple column [while updating more than one column, the column must be separated by comma operator]

Example: SQL> update department set loc='Hyderabad', dname= ‘cse’ where deptno=20;

Output:

1 row updated.

4. Selection of Records [Retrieving (Displaying) Data:]

Syntax:

Select <field1, field2 …fieldn> from <table name> where <condition>;

Example:

a) SQL> select * from department;

Output:

DEPTNO	DNAME	LOC
10	accounts	chennai
20	finance	Hyderabad
30	IT	Bangalore
40	marketing	chennai

Example:

b) SQL> select dname, loc from department;

Output:

DNAME	LOC
accounts	chennai
finance	Hyderabad
IT	Bangalore
marketing	Chennai

- **Using Alias name for a field**

Syntax:

Select <col1> <alias name 1> , <col2> < alias name 2> from < tab1>;

Example:

SQL> select dname, loc as location from department;

Output:

DNAME	LOCATION
accounts	chennai
finance	Hyderabad
IT	Bangalore
marketing	chennai

- **With distinct clause [Used to retrieve unique value from the column]**

Syntax:

Select distinct <col2> from < tab1>;

Example:

SQL> select distinct loc from department;

Output:

LOC
---
chennai
Bangalore
Hyderabad

- **Creating Table using subquery**

Syntax:

Create table <new _table_name> as Select <column names> from <old_table_name>;

Example:

SQL> create table copyOfEmp as select * from employee;

Output:

Table created.

- **To view the contents of new Table**

SQL> select * from copyofemp;

Output:

EMPLOYEE_ID FIRST_NAME LAST_NAME MANAGER HIRE_DATE JOB_ID SALARY COMMISION DEPTNO

111 Smith Ford 222 21-JUL-10 J1 30000 0.1 1 0

- **To create a table with same structure as an existing table**

Syntax:

Create table <new _table_name> as Select <column names> from<old_table_name>

where 1=2;

Example:

create table copyOfEmp2 as select * from employee where 1=2;

Output:

Table created.

SQL> select * from copyofemp2;

Output:

no rows selected

SQL> desc copyofemp2;

Output:

Name	Null?	Type
EMPLOYEE_ID		NUMBER(3)
FIRST_NAME		VARCHAR2(10)
LAST_NAME		VARCHAR2(10)
MANAGER		NUMBER(4)
HIRE_DATE		DATE
JOB_ID		VARCHAR2(10)
SALARY		NUMBER(10)
COMMISION		NUMBER(8)
DEPTNO		NUMBER(2)

Note: Only structure of table alone is copied and not the contents.

- **Inserting into table using a subquery**

Syntax :

Insert into <new_table_name> (Select <columnnames> from <old_table_name>);

Example:

SQL> insert into copyofemp2 (select * from employee where employee_id > 100);

Output:

50 rows created.